

Milk Hill 2001 - Complete 409-Circle Physical Reconstruction Register

All 409 circles are accounted for. Geometry-derived entries are part of the completed register and are not missing circles.

Founder: Lawrence Herber | Geometric Intelligence Engine (GIE) | Revision: 14 August 2026

Register fields: circle ID, arm, radial slot, provenance status, center radius, angle, ground coordinates, diameter, spacing and edge gap. CENTER is the 72-ft control circle. DERIVED_COMPLETE identifies geometry-derived entries retained for audit without treating the formation as incomplete.

Circle	Arm	Slot	Status	r ft	Theta deg	X ft	Y ft	d ft	d in	Spacing	Gap
CENTER	-1	0	CONTROL_CENTER	0.0	0.0	0.0	0.0	72.0	864.0		
A0-R01	0	1	MEASURED	37.8845	85.641326	2.8792	37.775	3.8222	45.87		
A0-R02	0	2	MEASURED	41.2802	90.629756	-0.4537	41.2777	4.2734	51.28	4.835	0.7872
A0-R03	0	3	MEASURED	42.2645	73.600016	11.933	40.5449	3.8222	45.87	12.4084	8.3606
A0-R04	0	4	MEASURED	44.5567	96.162986	-4.7835	44.2991	3.3333	40.0	17.1329	13.5551
A0-R05	0	5	DERIVED_COMPLETE	54.1261	89.040634	0.9063	54.1185	11.9909	143.89	11.3487	3.6866
A0-R06	0	6	MEASURED	64.6947	81.918282	9.0951	64.0522	20.6717	248.06	12.8739	-3.4574
A0-R07	0	7	MEASURED	68.7188	112.881534	-26.7197	63.3114	5.7333	68.8	35.8225	22.62
A0-R08	0	8	MEASURED	88.2175	93.256036	-5.0106	88.0751	3.3333	40.0	32.9322	28.3989
A0-R09	0	9	MEASURED	88.4359	97.165919	-11.0318	87.7451	4.6812	56.17	6.0302	2.023
A0-R10	0	10	MEASURED	90.6765	81.217237	13.8453	89.6132	30.8156	369.79	24.9471	7.1987
A0-R11	0	11	MEASURED	94.7401	70.465683	31.6784	89.287	3.8222	45.87	17.8361	0.5172
A0-R12	0	12	MEASURED	106.7312	114.128838	-43.6306	97.406	3.3333	40.0	75.7454	72.1677
A0-R13	0	13	MEASURED	122.7159	83.811524	13.2287	122.0008	35.5486	426.58	61.9507	42.5097
A0-R14	0	14	MEASURED	127.3118	93.808347	-8.456	127.0307	3.8222	45.87	22.2604	2.575
A0-R15	0	15	DERIVED_COMPLETE	131.0401	85.363344	10.5928	130.6113	4.9328	59.19	19.3824	15.0049
A0-R16	0	16	MEASURED	134.0526	75.448797	33.6801	129.7527	6.0434	72.52	23.1032	17.6151
A0-R17	0	17	MEASURED	136.1554	73.545074	38.5675	130.5788	6.0434	72.52	4.9568	-1.0866
A0-R18	0	18	MEASURED	149.8045	97.939521	-20.6922	148.3685	4.6812	56.17	61.8723	56.51
A0-R19	0	19	MEASURED	149.999	99.817764	-25.5771	147.8023	4.2734	51.28	4.9176	0.4403
A0-R20	0	20	MEASURED	154.9434	88.990303	2.7304	154.9193	42.131	505.57	29.1884	5.9862
A0-R21	0	21	MEASURED	164.7949	58.17513	86.9004	140.0203	4.6812	56.17	85.4785	62.0724
A0-R22	0	22	MEASURED	167.8065	85.502334	13.1591	167.2898	3.8222	45.87	78.6219	74.3702
A0-R23	0	23	MEASURED	179.7398	103.546684	-42.1018	174.7393	6.0434	72.52	55.7608	50.828
A0-R24	0	24	MEASURED	181.4606	106.835024	-52.5541	173.6837	8.5467	102.56	10.5054	3.2104
A0-R25	0	25	DERIVED_COMPLETE	186.7166	104.035493	-45.2831	181.1423	27.263	327.16	10.4163	-7.4886
A0-R26	0	26	MEASURED	192.2706	96.269827	-20.998	191.1206	43.9554	527.46	26.255	-9.3542
A0-R27	0	27	MEASURED	194.5467	62.697905	89.2351	172.8743	4.2734	51.28	111.733	87.6186
A0-R28	0	28	MEASURED	202.5952	61.072428	97.996	177.3178	9.3625	112.35	9.8234	3.0054
A0-R29	0	29	MEASURED	204.3379	110.998781	-73.2241	190.7674	6.6203	79.44	171.7476	163.7562
A0-R30	0	30	MEASURED	206.4785	59.023984	106.2702	177.0311	9.9304	119.16	180.0191	171.7438
A0-R31	0	31	MEASURED	206.8868	113.315763	-81.8854	189.9919	7.8797	94.56	188.6015	179.6964
A0-R32	0	32	MEASURED	211.3992	69.998286	72.3087	198.6481	4.2734	51.28	154.4369	148.3604
A0-R33	0	33	MEASURED	219.0475	92.247309	-8.5895	218.8791	11.6248	139.5	83.3895	75.4404
A0-R34	0	34	MEASURED	219.7304	68.446812	80.7212	204.3662	6.3384	76.06	90.4822	81.5006
A0-R35	0	35	DERIVED_COMPLETE	222.4347	66.968128	87.026	204.7038	5.877	70.52	6.3139	0.2062

Circle	Arm	Slot	Status	r ft	Theta deg	X ft	Y ft	d ft	d in	Spacing	Gap
A0-R36	0	36	MEASURED	225.4503	66.520714	89.8234	206.784	5.7333	68.8	3.486	-2.3192
A0-R37	0	37	MEASURED	226.6879	104.475704	-56.665	219.4914	44.1212	529.45	147.0385	122.1113
A0-R38	0	38	MEASURED	231.5069	78.349226	46.7519	226.7371	3.3333	40.0	103.6704	79.9432
A0-R39	0	39	MEASURED	232.3618	92.858392	-11.5873	232.0727	3.8222	45.87	58.5827	55.0049
A0-R40	0	40	MEASURED	244.1196	84.680814	22.6309	243.0683	6.3384	76.06	35.9415	30.8612
A0-R41	0	41	MEASURED	244.3905	77.528666	52.7764	238.6239	8.1081	97.3	30.4714	23.2482
A0-R42	0	42	MEASURED	246.1704	75.887128	60.0244	238.7403	9.9304	119.16	7.249	-1.7703
A0-R43	0	43	MEASURED	250.1371	60.946309	121.4738	218.6611	62.4554	749.46	64.6467	28.4538
A0-R44	0	44	DERIVED_COMPLETE	255.1322	88.196603	8.029	255.0058	60.8593	730.31	119.1246	57.4672
A0-R45	0	45	MEASURED	258.3806	115.446897	-111.0195	233.3135	55.816	669.79	121.0087	62.6711
A0-R46	0	46	MEASURED	258.6995	83.681463	28.4714	257.128	6.0434	72.52	141.5092	110.5795
A0-R47	0	47	MEASURED	261.3959	103.031989	-58.9435	254.6634	3.8222	45.87	87.4496	82.5168
A0-R48	0	48	MEASURED	268.2556	114.024438	-109.2139	245.0171	3.3333	40.0	51.1876	47.6098
A0-R49	0	49	MEASURED	270.4106	106.696012	-77.6873	259.0108	9.7448	116.94	34.4927	27.9537
A0-R50	0	50	MEASURED	271.5797	74.773261	71.3276	262.0456	35.0832	421.0	149.0458	126.6318
A0-R51	0	51	MEASURED	272.9311	108.536111	-86.7654	258.7724	11.3062	135.67	158.1269	134.9322
A0-R52	0	52	MEASURED	274.8144	69.045455	98.2811	256.6394	32.3197	387.84	185.0588	163.2459
A0-R53	0	53	MEASURED	278.5026	102.997302	-62.6367	271.3676	28.0223	336.27	161.5904	131.4194
A0-R54	0	54	DERIVED_COMPLETE	279.0452	100.990003	-53.1965	273.9277	17.4783	209.74	9.7811	-12.9692
A0-R55	0	55	MEASURED	279.6181	95.923765	-28.858	278.125	9.7448	116.94	24.6978	11.0863
A0-R56	0	56	MEASURED	285.2114	81.223176	43.5193	281.8717	41.3435	496.12	72.4742	46.9301
A0-R57	0	57	MEASURED	285.7514	60.691491	139.8787	249.1743	6.8906	82.69	101.7559	77.6388
A0-R58	0	58	MEASURED	286.144	98.683884	-43.2028	282.8637	19.583	235.0	186.1554	172.9186
A0-R59	0	59	MEASURED	286.7212	92.542768	-12.7204	286.4389	9.5555	114.67	30.6913	16.1221
A0-R60	0	60	MEASURED	288.5399	115.477352	-124.1167	260.4809	8.9639	107.57	114.3807	105.121
A0-R61	0	61	MEASURED	288.7629	88.017842	9.9878	288.5901	24.6969	296.36	137.0188	120.1884
A0-R62	0	62	MEASURED	289.9604	58.934445	149.6249	248.3736	7.4017	88.82	145.3131	129.2638
A0-R63	0	63	MEASURED	295.2954	103.34184	-68.1425	287.3255	3.8222	45.87	221.2236	215.6117
A0-R64	0	64	DERIVED_COMPLETE	295.7597	104.020838	-71.6551	286.9484	4.155	49.86	3.5329	-0.4557
A0-R65	0	65	MEASURED	296.2428	104.214772	-72.7446	287.1725	5.0563	60.68	1.1123	-3.4934
A0-R66	0	66	MEASURED	304.7728	96.512709	-34.5684	302.8061	8.5467	102.56	41.2532	34.4517
A0-R67	0	67	MEASURED	304.8741	72.353096	92.4226	290.5276	4.6812	56.17	127.5833	120.9693
A0-R68	0	68	MEASURED	312.9661	80.908308	49.4533	309.0342	5.0563	60.68	46.7852	41.9165
A1-R01	1	1	MEASURED	23.737	163.095385	-22.7113	6.9022	3.3333	40.0		
A1-R02	1	2	MEASURED	31.5956	118.189401	-14.9254	27.848	51.1273	613.53	22.3461	-4.8842
A1-R03	1	3	MEASURED	53.6134	155.148626	-48.6489	22.5319	11.7632	141.16	34.14	2.6947
A1-R04	1	4	MEASURED	63.9936	160.65264	-60.3798	21.2007	11.6558	139.87	11.8062	0.0967
A1-R05	1	5	DERIVED_COMPLETE	69.1974	141.348977	-54.0407	43.219	11.6122	139.35	22.9126	11.2786
A1-R06	1	6	MEASURED	72.0312	122.045314	-38.219	61.0557	11.5474	138.57	23.8427	12.2629
A1-R07	1	7	MEASURED	72.9153	169.614521	-71.7207	13.1444	9.9055	118.87	58.4624	47.736
A1-R08	1	8	MEASURED	86.8989	137.78762	-64.3625	58.3857	4.1966	50.36	45.8357	38.7847
A1-R09	1	9	MEASURED	89.3441	120.347207	-45.1401	77.1021	11.7632	141.16	26.8292	18.8493
A1-R10	1	10	MEASURED	90.2338	173.06647	-89.5739	10.8928	13.0797	156.96	79.7373	67.3158
A1-R11	1	11	MEASURED	92.6477	158.723713	-86.333	33.6187	4.4863	53.84	22.9558	14.1728
A1-R12	1	12	MEASURED	96.4733	128.767251	-60.4075	75.2198	9.3838	112.61	49.0182	42.0832

Circle	Arm	Slot	Status	r ft	Theta deg	X ft	Y ft	d ft	d in	Spacing	Gap
A1-R13	1	13	DERIVED_COMPLETE	96.7196	144.640457	-78.8784	55.9722	6.0655	72.79	26.6767	18.952
A1-R14	1	14	MEASURED	96.9831	160.513663	-91.428	32.3518	3.3333	40.0	26.7472	22.0478
A1-R15	1	15	MEASURED	115.9476	128.248439	-71.78	91.0576	11.7632	141.16	61.9065	54.3582
A1-R16	1	16	MEASURED	118.4939	144.123701	-96.0138	69.4419	3.5467	42.56	32.4733	24.8184
A1-R17	1	17	MEASURED	125.125	173.940318	-124.4259	13.2087	7.6069	91.28	63.0033	57.4265
A1-R18	1	18	MEASURED	130.7187	135.164005	-92.6963	92.1671	17.6626	211.95	85.0952	72.4605
A1-R19	1	19	MEASURED	140.2498	122.730418	-75.8312	117.9814	3.5467	42.56	30.8352	20.2305
A1-R20	1	20	MEASURED	148.2025	136.59625	-107.6735	101.8351	19.2964	231.56	35.702	24.2805
A1-R21	1	21	MEASURED	148.4843	149.74835	-128.2639	74.8062	3.3333	40.0	33.9783	22.6634
A1-R22	1	22	DERIVED_COMPLETE	154.2534	147.193316	-129.6505	83.5754	12.7175	152.61	8.8781	0.8527
A1-R23	1	23	MEASURED	165.2693	141.876365	-130.0142	102.0308	22.2627	267.15	18.4589	0.9688
A1-R24	1	24	MEASURED	179.869	133.720299	-124.3144	129.9953	3.3333	40.0	28.5395	15.7415
A1-R25	1	25	MEASURED	182.7159	158.464423	-169.9604	67.0711	7.0935	85.12	77.7368	72.5234
A1-R26	1	26	MEASURED	191.6478	159.892155	-179.9663	65.8863	7.6069	91.28	10.0757	2.7255
A1-R27	1	27	MEASURED	193.1595	148.371749	-164.4692	101.2939	37.0291	444.35	38.6505	16.3325
A1-R28	1	28	MEASURED	205.0213	167.405274	-200.0878	44.7056	3.5467	42.56	66.865	46.5771
A1-R29	1	29	MEASURED	213.9457	168.392225	-209.5701	43.0482	3.8853	46.62	9.626	5.91
A1-R30	1	30	DERIVED_COMPLETE	214.0977	155.753832	-195.2121	87.9209	5.6197	67.44	47.1138	42.3613
A1-R31	1	31	MEASURED	214.2484	143.115438	-171.3658	128.5929	7.0935	85.12	47.1472	40.7906
A1-R32	1	32	MEASURED	220.8185	144.606273	-180.0093	127.8963	7.0935	85.12	8.6715	1.578
A1-R33	1	33	MEASURED	222.7148	117.122413	-101.5341	198.2238	6.7295	80.75	105.377	98.4655
A1-R34	1	34	MEASURED	228.3357	159.22625	-213.4912	80.9858	43.3226	519.87	162.1085	137.0824
A1-R35	1	35	MEASURED	238.6648	167.726494	-233.2099	50.735	3.5467	42.56	36.11	12.6754
A1-R36	1	36	MEASURED	238.7019	156.153254	-218.3239	96.5052	3.3333	40.0	48.13	44.69
A1-R37	1	37	MEASURED	254.1257	132.391886	-171.331	187.6848	3.3333	40.0	102.5771	99.2438
A1-R38	1	38	MEASURED	257.0189	139.146393	-194.4048	168.1234	3.5467	42.56	30.2498	26.8098
A1-R39	1	39	DERIVED_COMPLETE	258.4613	136.56627	-187.6869	177.6961	4.2654	51.18	11.6947	7.7887
A1-R40	1	40	MEASURED	259.712	133.986148	-180.3659	186.8648	4.7585	57.1	11.7329	7.221
A1-R41	1	41	MEASURED	261.4334	159.028579	-244.1158	93.5676	3.5467	42.56	112.9974	108.8448
A1-R42	1	42	MEASURED	264.071	140.796755	-204.6309	166.9122	3.3333	40.0	83.2976	79.8576
A1-R43	1	43	MEASURED	264.8995	171.899696	-262.2566	37.3261	54.1619	649.94	141.8213	113.0737
A1-R44	1	44	MEASURED	265.9583	159.798484	-249.5975	91.8415	3.5467	42.56	55.9659	27.1116
A1-R45	1	45	MEASURED	266.1174	154.146594	-239.4824	116.0458	4.4863	53.84	26.2329	22.2164
A1-R46	1	46	MEASURED	268.0471	124.031507	-150.0122	222.1387	25.3784	304.54	138.7826	123.8502
A1-R47	1	47	DERIVED_COMPLETE	268.9454	136.649896	-195.5698	184.6188	14.9324	179.19	59.0189	38.8635
A1-R48	1	48	MEASURED	269.9841	154.904768	-244.4988	114.5068	4.4863	53.84	85.497	75.7877
A1-R49	1	49	MEASURED	273.4898	164.176968	-263.1268	74.5716	5.719	68.63	44.0661	38.9634
A1-R50	1	50	MEASURED	277.7216	134.952993	-196.2177	196.5399	20.5589	246.71	139.1154	125.9764
A1-R51	1	51	MEASURED	281.5516	144.698143	-229.7796	162.7042	17.3029	207.63	47.6577	28.7268
A1-R52	1	52	MEASURED	282.2162	164.753609	-272.2833	74.2145	8.3931	100.72	98.1682	85.3202
A1-R53	1	53	MEASURED	285.1496	159.794596	-267.6016	98.4869	24.4185	293.02	24.7197	8.3139
A1-R54	1	54	MEASURED	286.6234	128.932407	-180.115	222.9609	24.9787	299.74	152.1436	127.445
A1-R55	1	55	MEASURED	287.9319	123.604995	-159.36	239.8107	3.3333	40.0	26.7337	12.5777
A1-R56	1	56	DERIVED_COMPLETE	289.5861	138.613163	-217.2657	191.4568	15.8676	190.41	75.4399	65.8394
A1-R57	1	57	MEASURED	290.6854	153.621332	-260.4184	129.152	28.988	347.86	75.7895	53.3617

Circle	Arm	Slot	Status	r ft	Theta deg	X ft	Y ft	d ft	d in	Spacing	Gap
A1-R58	1	58	MEASURED	290.8223	117.018614	-132.1147	259.0817	5.2607	63.13	182.6022	165.4778
A1-R59	1	59	MEASURED	292.9559	138.186021	-218.3439	195.3179	3.3333	40.0	107.2441	102.9471
A1-R60	1	60	MEASURED	298.222	139.992541	-228.4263	191.7231	15.7021	188.43	10.7041	1.1864
A1-R61	1	61	MEASURED	300.4543	147.045264	-252.1114	163.44	11.8697	142.44	36.8906	23.1047
A1-R62	1	62	MEASURED	303.6448	172.382156	-300.9649	40.2527	11.8697	142.44	132.5209	120.6512
A1-R63	1	63	MEASURED	305.7954	125.196369	-176.2545	249.8903	6.5399	78.48	243.9275	234.7227
A1-R64	1	64	DERIVED_COMPLETE	308.8098	139.410861	-234.5085	200.921	4.6432	55.72	76.1021	70.5105
A1-R65	1	65	MEASURED	310.7763	153.625353	-278.4271	138.0589	3.8853	46.62	76.6843	72.4201
A1-R66	1	66	MEASURED	310.8876	126.208797	-183.6505	250.8458	3.8853	46.62	147.321	143.4357
A1-R67	1	67	MEASURED	312.3369	144.442532	-254.0963	181.6299	3.8853	46.62	98.7595	94.8742
A1-R68	1	68	MEASURED	327.6121	136.900572	-239.2122	223.8463	3.3333	40.0	44.7634	41.1541
A2-R01	2	1	MEASURED	31.0318	207.073466	-27.6314	-14.1236	15.134	181.61		
A2-R02	2	2	MEASURED	55.1223	232.843772	-33.2933	-43.932	27.1571	325.89	30.3414	9.1958
A2-R03	2	3	MEASURED	63.257	183.327357	-63.1503	-3.6715	3.3333	40.0	50.1234	34.8782
A2-R04	2	4	MEASURED	68.2781	216.723744	-54.7268	-40.8274	5.8614	70.34	38.0988	33.5014
A2-R05	2	5	DERIVED_COMPLETE	68.3557	200.271551	-64.1218	-23.6832	4.4441	53.33	19.5497	14.3969
A2-R06	2	6	MEASURED	68.4344	183.819357	-68.2824	-4.5585	3.3333	40.0	19.5721	15.6834
A2-R07	2	7	MEASURED	89.9886	193.452746	-87.5195	-20.9352	3.3841	40.61	25.2639	21.9052
A2-R08	2	8	MEASURED	94.4013	193.904356	-91.6351	-22.6848	3.3333	40.0	4.4721	1.1134
A2-R09	2	9	MEASURED	99.4945	233.569246	-59.0849	-80.0508	5.4567	65.48	65.9574	61.5624
A2-R10	2	10	MEASURED	99.8358	183.970043	-99.5962	-6.9121	20.9157	250.99	83.6088	70.4226
A2-R11	2	11	MEASURED	118.2316	182.744213	-118.096	-5.6606	20.3045	243.65	18.542	-2.0681
A2-R12	2	12	MEASURED	121.3028	202.148423	-112.3519	-45.732	3.3333	40.0	40.481	28.6621
A2-R13	2	13	DERIVED_COMPLETE	125.8026	198.714682	-119.1512	-40.3645	13.2272	158.73	8.6626	0.3824
A2-R14	2	14	MEASURED	131.1817	195.280941	-126.5438	-34.5732	23.8331	286.0	9.3909	-9.1392
A2-R15	2	15	MEASURED	137.1233	216.478343	-110.2583	-81.5224	5.6626	67.95	49.6935	34.9456
A2-R16	2	16	MEASURED	153.2891	214.531457	-126.2819	-86.8932	17.0552	204.66	16.8997	5.5408
A2-R17	2	17	MEASURED	154.5831	193.736675	-150.1616	-36.7073	24.3089	291.71	55.5776	34.8956
A2-R18	2	18	MEASURED	157.3747	186.129622	-156.475	-16.8042	3.7071	44.49	20.8804	6.8724
A2-R19	2	19	MEASURED	163.4719	185.914007	-162.6019	-16.8434	4.5402	54.48	6.127	2.0034
A2-R20	2	20	MEASURED	171.9742	227.241152	-116.7557	-126.2665	4.0041	48.05	118.6393	114.3671
A2-R21	2	21	MEASURED	179.4895	205.644709	-161.8089	-77.6811	48.2394	578.87	66.2595	40.1378
A2-R22	2	22	DERIVED_COMPLETE	181.0507	206.890903	-161.4735	-81.888	26.1217	313.46	4.2202	-32.9604
A2-R23	2	23	MEASURED	182.609	210.368188	-157.554	-92.3189	4.0041	48.05	11.143	-3.9199
A2-R24	2	24	MEASURED	190.051	225.834444	-132.415	-136.3292	16.0164	192.2	50.6842	40.6739
A2-R25	2	25	MEASURED	190.1162	210.62857	-163.5927	-96.8586	3.3333	40.0	50.2989	40.6241
A2-R26	2	26	MEASURED	206.1326	200.674211	-192.8583	-72.7759	7.0985	85.18	37.9005	32.6846
A2-R27	2	27	MEASURED	217.6794	200.560684	-203.8134	-76.4489	10.4852	125.82	11.5544	2.7626
A2-R28	2	28	MEASURED	221.2206	180.247942	-221.2185	-0.9573	3.3333	40.0	77.472	70.5628
A2-R29	2	29	MEASURED	224.4013	219.606191	-172.8887	-143.0575	70.043	840.52	150.0941	113.4059
A2-R30	2	30	DERIVED_COMPLETE	228.1008	199.09446	-215.5509	-74.6178	37.6428	451.71	80.6476	26.8047
A2-R31	2	31	MEASURED	230.5768	178.582728	-230.5063	5.703	5.2426	62.91	81.7013	60.2586
A2-R32	2	32	MEASURED	231.0035	200.459221	-216.432	-80.7451	7.258	87.1	87.5863	81.336
A2-R33	2	33	MEASURED	239.624	178.643503	-239.5569	5.6726	5.6626	67.95	89.4583	82.998
A2-R34	2	34	MEASURED	250.3961	194.124225	-242.8264	-61.1029	7.0985	85.18	66.8556	60.475

Circle	Arm	Slot	Status	r ft	Theta deg	X ft	Y ft	d ft	d in	Spacing	Gap
A2-R35	2	35	MEASURED	251.7948	188.080614	-249.2948	-35.3939	6.9353	83.22	26.5103	19.4934
A2-R36	2	36	MEASURED	259.4625	188.011079	-256.9304	-36.1599	11.0178	132.21	7.6739	-1.3026
A2-R37	2	37	MEASURED	267.2671	195.529322	-257.5102	-71.5558	14.0347	168.42	35.4007	22.8744
A2-R38	2	38	MEASURED	270.59	217.164577	-215.6341	-163.4652	5.2426	62.91	100.9997	91.361
A2-R39	2	39	DERIVED_COMPLETE	274.6155	217.044669	-219.1888	-165.4387	6.1392	73.67	4.0658	-1.6251
A2-R40	2	40	MEASURED	277.1661	216.632936	-222.4188	-165.3812	7.567	90.8	3.2305	-3.6226
A2-R41	2	41	MEASURED	277.3747	203.208579	-254.9285	-109.3077	8.5611	102.73	64.8161	56.752
A2-R42	2	42	MEASURED	280.7387	210.647141	-241.526	-143.1064	5.8614	70.34	36.3591	29.1478
A2-R43	2	43	MEASURED	283.4093	179.517691	-283.3992	2.3857	3.3333	40.0	151.3979	146.8005
A2-R44	2	44	MEASURED	291.3487	224.254286	-208.6784	-203.316	27.3252	327.9	218.8524	203.5231
A2-R45	2	45	MEASURED	292.6975	184.320983	-291.8655	-22.053	54.4405	653.29	199.4401	158.5573
A2-R46	2	46	MEASURED	294.7728	194.106248	-285.884	-71.8423	40.1266	481.52	50.1473	2.8637
A2-R47	2	47	DERIVED_COMPLETE	295.745	191.487649	-289.8206	-58.8996	19.4417	233.3	13.5281	-16.2561
A2-R48	2	48	MEASURED	296.8479	188.86905	-293.2986	-45.767	5.2426	62.91	13.5853	1.2432
A2-R49	2	49	MEASURED	300.3066	236.282732	-166.6987	-249.7911	3.7071	44.49	240.1111	235.6362
A2-R50	2	50	MEASURED	301.313	218.537033	-235.6888	-187.7242	35.4925	425.91	92.8005	73.2007
A2-R51	2	51	MEASURED	306.9208	198.514469	-291.0356	-97.4609	9.6905	116.29	105.8808	83.2893
A2-R52	2	52	MEASURED	308.3276	211.771605	-262.1259	-162.3451	42.4293	509.15	71.0334	44.9735
A2-R53	2	53	MEASURED	310.297	204.003449	-283.4628	-126.2262	44.2525	531.03	41.9505	-1.3904
A2-R54	2	54	MEASURED	316.4431	207.562618	-280.5286	-146.4238	3.3333	40.0	20.4096	-3.3833
A2-R55	2	55	MEASURED	316.9352	214.251555	-261.9706	-178.3798	3.3333	40.0	36.9538	33.6205
A2-R56	2	56	DERIVED_COMPLETE	318.7571	224.963471	-225.539	-225.2516	8.6471	103.77	59.3652	53.375
A2-R57	2	57	MEASURED	320.3285	231.925819	-197.5406	-252.1668	19.2625	231.15	38.8373	24.8825
A2-R58	2	58	MEASURED	320.4571	195.281926	-309.1259	-84.4625	29.2679	351.21	201.4349	177.1697
A2-R59	2	59	MEASURED	327.2853	220.97247	-247.1085	-214.5998	12.4799	149.76	144.1592	123.2853
A2-R60	2	60	MEASURED	329.4996	182.932909	-329.068	-16.8594	14.1161	169.39	214.0529	200.7549
A2-R61	2	61	MEASURED	338.6269	213.408673	-282.6737	-186.4504	10.9133	130.96	175.8225	163.3078
A2-R62	2	62	MEASURED	340.3854	233.449468	-202.7102	-273.4424	4.5402	54.48	118.16	110.4332
A2-R63	2	63	MEASURED	342.0852	206.189476	-306.9665	-150.9762	11.5257	138.31	160.8333	152.8004
A2-R64	2	64	DERIVED_COMPLETE	343.7864	200.534807	-321.942	-120.5922	16.0143	192.17	33.8741	20.1041
A2-R65	2	65	MEASURED	345.7137	195.843119	-332.581	-94.3813	18.1609	217.93	28.2877	11.2001
A2-R66	2	66	MEASURED	348.366	184.206292	-347.4277	-25.5519	5.2426	62.91	70.4125	58.7108
A2-R67	2	67	MEASURED	349.5257	235.049151	-200.234	-286.4866	3.3333	40.0	299.5879	295.2999
A2-R68	2	68	MEASURED	365.2614	197.45415	-348.4437	-109.5574	7.4141	88.97	230.803	225.4293
A3-R01	3	1	MEASURED	46.5828	254.297633	-12.6072	-44.8444	7.4536	89.44		
A3-R02	3	2	MEASURED	55.2968	292.855432	21.4777	-50.9554	10.8866	130.64	34.6284	25.4583
A3-R03	3	3	MEASURED	76.5286	265.061129	-6.5886	-76.2444	10.1835	122.2	37.779	27.2439
A3-R04	3	4	DERIVED_COMPLETE	80.7219	253.20718	-23.3215	-77.2796	28.1456	337.75	16.7649	-2.3996
A3-R05	3	5	MEASURED	85.0108	248.223314	-31.5382	-78.9442	46.1078	553.29	8.3836	-28.7431
A3-R06	3	6	MEASURED	109.1953	273.602819	6.8618	-108.9795	7.2008	86.41	48.7512	22.0969
A3-R07	3	7	MEASURED	122.5174	260.484011	-20.2549	-120.8315	53.472	641.66	29.5937	-0.7427
A3-R08	3	8	MEASURED	125.327	288.480411	39.7262	-118.8642	9.6225	115.47	60.0134	28.4661
A3-R09	3	9	MEASURED	141.3032	283.692032	33.4469	-137.2877	23.2538	279.05	19.4642	3.0261
A3-R10	3	10	DERIVED_COMPLETE	143.7124	265.514107	-11.2403	-143.2721	17.828	213.94	45.0861	24.5452
A3-R11	3	11	MEASURED	145.3466	250.830938	-47.7255	-137.2877	9.2296	110.76	36.9728	23.444

Circle	Arm	Slot	Status	r ft	Theta deg	X ft	Y ft	d ft	d in	Spacing	Gap
A3-R12	3	12	MEASURED	148.1153	278.668265	22.3229	-146.4234	3.3333	40.0	70.6417	64.3603
A3-R13	3	13	MEASURED	151.6642	279.66586	25.4647	-149.5112	3.849	46.19	4.4051	0.814
A3-R14	3	14	MEASURED	167.4034	271.868837	5.4593	-167.3143	69.7084	836.5	26.78	-9.9987
A3-R15	3	15	MEASURED	182.4432	295.714159	79.1588	-164.3759	22.2777	267.33	73.7581	27.765
A3-R16	3	16	DERIVED_COMPLETE	186.5176	284.371319	46.2946	-180.681	13.6515	163.82	36.6867	18.7221
A3-R17	3	17	MEASURED	188.5782	273.028479	9.963	-188.3148	8.6066	103.28	37.1249	25.9959
A3-R18	3	18	MEASURED	189.6181	294.512868	78.6721	-172.5275	3.3333	40.0	70.4995	64.5295
A3-R19	3	19	MEASURED	189.8773	289.689275	63.9733	-178.7759	3.3333	40.0	15.9718	12.6385
A3-R20	3	20	MEASURED	195.8556	291.049866	70.3475	-182.7858	5.0918	61.1	7.5306	3.3181
A3-R21	3	21	MEASURED	199.0942	264.301241	-19.7697	-198.1103	21.8581	262.3	91.4109	77.9359
A3-R22	3	22	DERIVED_COMPLETE	201.2071	246.820848	-79.1966	-184.9654	16.5399	198.48	60.8633	41.6643
A3-R23	3	23	MEASURED	203.8219	238.413839	-106.7579	-173.6264	11.2217	134.66	29.8026	15.9218
A3-R24	3	24	MEASURED	211.1165	284.554175	53.0526	-204.3419	72.0	864.0	162.7354	121.1246
A3-R25	3	25	MEASURED	218.3911	263.888096	-23.2522	-217.1497	5.7735	69.28	77.3723	38.4855
A3-R26	3	26	MEASURED	223.3151	238.716435	-115.9617	-190.8468	24.7207	296.65	96.3685	81.1214
A3-R27	3	27	MEASURED	228.4656	247.48273	-87.4936	-211.0483	12.9099	154.92	34.9075	16.0922
A3-R28	3	28	DERIVED_COMPLETE	236.862	255.381067	-59.7814	-229.1938	13.1044	157.25	33.1244	20.1172
A3-R29	3	29	MEASURED	245.02	259.619448	-44.149	-241.0097	13.7437	164.92	19.5956	6.1715
A3-R30	3	30	MEASURED	249.5981	248.912216	-89.8049	-232.8826	23.3333	280.0	46.3736	27.8351
A3-R31	3	31	MEASURED	262.4389	296.920674	118.8209	-233.9995	3.849	46.19	208.6288	195.0376
A3-R32	3	32	MEASURED	264.0133	282.300034	56.243	-257.953	9.0267	108.32	67.0057	60.5678
A3-R33	3	33	MEASURED	264.7286	246.806397	-104.2605	-243.3331	5.4433	65.32	161.168	153.933
A3-R34	3	34	MEASURED	265.2436	283.469168	61.7811	-257.9482	6.9389	83.27	166.6836	160.4925
A3-R35	3	35	DERIVED_COMPLETE	266.819	274.59542	21.3773	-265.9612	16.271	195.25	41.1907	29.5857
A3-R36	3	36	MEASURED	268.0819	260.005433	-46.5269	-264.0135	24.4192	293.03	67.9322	47.5871
A3-R37	3	37	MEASURED	268.1353	248.801685	-96.957	-249.9918	10.5409	126.49	52.3431	34.8631
A3-R38	3	38	MEASURED	269.5871	244.420107	-116.3994	-243.1633	8.8192	105.83	20.6067	10.9267
A3-R39	3	39	MEASURED	272.036	295.893591	118.7984	-244.7254	4.714	56.57	235.203	228.4364
A3-R40	3	40	MEASURED	272.2065	294.092462	111.1175	-248.494	4.3033	51.64	8.5557	4.047
A3-R41	3	41	DERIVED_COMPLETE	272.4653	293.439507	108.3814	-249.9816	5.1946	62.34	3.1143	-1.6346
A3-R42	3	42	MEASURED	272.8871	292.481627	104.3485	-252.1483	6.0858	73.03	4.5781	-1.0621
A3-R43	3	43	MEASURED	273.9396	247.77239	-103.6278	-253.5827	3.849	46.19	207.9812	203.0138
A3-R44	3	44	MEASURED	274.5267	275.161585	24.6978	-273.4135	6.9389	83.27	129.8487	124.4548
A3-R45	3	45	MEASURED	274.7704	273.993516	19.136	-274.1032	11.2217	134.66	5.6044	-3.4759
A3-R46	3	46	MEASURED	276.1985	267.684937	-11.1569	-275.9731	12.7657	153.19	30.3505	18.3568
A3-R47	3	47	DERIVED_COMPLETE	277.9976	278.408006	40.6492	-275.0096	21.419	257.03	51.815	34.7227
A3-R48	3	48	MEASURED	281.2388	289.131075	92.1705	-265.7064	28.8675	346.41	52.3546	27.2113
A3-R49	3	49	MEASURED	295.656	285.378456	78.4061	-285.0701	25.2396	302.88	23.7574	-3.2962
A3-R50	3	50	MEASURED	297.8509	258.209533	-60.8608	-291.5667	4.3033	51.64	139.4184	124.6469
A3-R51	3	51	MEASURED	304.4033	294.753123	127.4565	-276.4348	13.7437	164.92	188.9243	179.9008
A3-R52	3	52	MEASURED	308.3034	296.475175	137.4448	-275.9709	10.1835	122.2	9.9991	-1.9645
A3-R53	3	53	DERIVED_COMPLETE	318.4423	296.472403	141.951	-285.0534	6.1727	74.07	10.1389	1.9608
A3-R54	3	54	MEASURED	326.1601	296.462512	145.3409	-291.987	3.849	46.19	7.718	2.7071
A3-R55	3	55	MEASURED	326.3637	284.152242	79.7957	-316.4584	15.396	184.75	69.9645	60.342
A3-R56	3	56	MEASURED	338.2326	268.890586	-6.5488	-338.1692	8.3887	100.66	89.0321	77.1398

Circle	Arm	Slot	Status	r ft	Theta deg	X ft	Y ft	d ft	d in	Spacing	Gap
A3-R57	3	57	MEASURED	338.8523	279.186207	54.0956	-334.5065	20.0924	241.11	60.7549	46.5143
A3-R58	3	58	MEASURED	344.3107	281.440387	68.2934	-337.4698	3.3333	40.0	14.5038	2.7909
A3-R59	3	59	DERIVED_COMPLETE	346.048	263.483786	-39.271	-343.8125	13.491	161.89	107.7513	99.3392
A3-R60	3	60	MEASURED	347.9761	245.527185	-144.1531	-316.7132	23.6487	283.78	108.3264	89.7566
A3-R61	3	61	MEASURED	356.9611	268.907544	-6.8057	-356.8962	19.7203	236.64	143.1047	121.4202
A3-R62	3	62	MEASURED	363.5025	257.728436	-77.2608	-355.1969	31.798	381.58	70.4755	44.7164
A3-R63	3	63	MEASURED	369.9728	270.269534	1.7404	-369.9687	5.4433	65.32	80.3704	61.7498
A3-R64	3	64	MEASURED	371.281	246.921733	-145.5378	-341.5675	12.3228	147.87	149.9917	141.1086
A3-R65	3	65	DERIVED_COMPLETE	373.82	247.349699	-143.9602	-344.9882	8.7073	104.49	3.767	-6.7481
A3-R66	3	66	MEASURED	377.8517	248.663254	-137.4809	-351.953	5.0918	61.1	9.5127	2.6131
A3-R67	3	67	MEASURED	385.3291	258.915964	-74.079	-378.1413	11.7063	140.48	68.5976	60.1985
A3-R68	3	68	MEASURED	391.8333	260.369556	-65.5508	-386.3113	3.3333	40.0	11.8101	4.2903
A4-R01	4	1	MEASURED	19.8539	316.663029	14.4403	-13.6255	19.7047	236.46		
A4-R02	4	2	MEASURED	50.308	341.311438	47.6555	-16.1199	4.6734	56.08	33.3087	21.1196
A4-R03	4	3	MEASURED	51.0281	317.708574	37.7471	-34.3369	27.161	325.93	20.7373	4.8201
A4-R04	4	4	DERIVED_COMPLETE	52.3762	330.77967	45.7113	-25.5684	16.0436	192.52	11.8454	-9.7569
A4-R05	4	5	MEASURED	55.4452	343.850765	53.2574	-15.4215	4.9262	59.11	12.6453	2.1604
A4-R06	4	6	MEASURED	80.1636	339.97712	75.3182	-27.4476	7.6316	91.58	25.1258	18.8469
A4-R07	4	7	MEASURED	81.2175	329.101286	69.6908	-41.707	3.3333	40.0	15.3296	9.8471
A4-R08	4	8	MEASURED	83.5314	332.529393	74.113	-38.5325	3.3333	40.0	5.4436	2.1103
A4-R09	4	9	MEASURED	86.3495	322.10808	68.1445	-53.0336	34.9723	419.67	15.6814	-3.4714
A4-R10	4	10	MEASURED	102.4081	308.015547	63.0706	-80.6815	9.7284	116.74	28.1097	5.7593
A4-R11	4	11	DERIVED_COMPLETE	114.1847	326.161364	94.8428	-63.5844	7.0746	84.9	36.0803	27.6788
A4-R12	4	12	MEASURED	122.0831	344.30718	117.5325	-33.021	6.2312	74.77	38.065	31.4121
A4-R13	4	13	MEASURED	125.4693	329.731991	108.3649	-63.2422	40.6819	488.18	31.5811	8.1246
A4-R14	4	14	MEASURED	127.8048	345.292198	123.6171	-32.4483	3.3333	40.0	34.3641	12.3565
A4-R15	4	15	MEASURED	139.373	336.77486	128.0785	-54.9611	5.6167	67.4	22.9506	18.4756
A4-R16	4	16	MEASURED	149.6101	352.573007	148.3549	-19.339	9.216	110.59	40.9886	33.5722
A4-R17	4	17	DERIVED_COMPLETE	150.9582	337.035985	138.9948	-58.8968	9.4157	112.99	40.6501	31.3342
A4-R18	4	18	MEASURED	152.2105	321.498962	119.1195	-94.7554	9.4757	113.71	40.9984	31.5527
A4-R19	4	19	MEASURED	158.6073	352.818072	157.3629	-19.8291	8.9488	107.39	84.122	74.9097
A4-R20	4	20	MEASURED	164.3487	299.313443	80.463	-143.3046	7.3067	87.68	145.464	137.3363
A4-R21	4	21	MEASURED	170.2938	340.46864	160.4948	-56.9331	49.8493	598.19	117.7503	89.1723
A4-R22	4	22	MEASURED	176.2605	346.922285	171.689	-39.8829	3.3333	40.0	20.3965	-6.1948
A4-R23	4	23	MEASURED	196.7266	351.113018	194.3649	-30.3915	37.9027	454.83	24.5822	3.9642
A4-R24	4	24	DERIVED_COMPLETE	197.751	345.241334	191.2266	-50.3767	22.1634	265.96	20.2301	-9.8029
A4-R25	4	25	MEASURED	198.4884	332.503985	176.0678	-91.6395	4.9262	59.11	43.9592	30.4144
A4-R26	4	26	MEASURED	200.6208	310.334636	129.852	-152.9287	3.3333	40.0	76.7611	72.6313
A4-R27	4	27	MEASURED	202.7345	311.283478	133.7612	-152.3458	3.3333	40.0	3.9524	0.6191
A4-R28	4	28	MEASURED	209.1572	332.909427	186.2101	-95.2498	5.8287	69.94	77.5295	72.9485
A4-R29	4	29	MEASURED	213.9359	317.604259	157.9928	-144.2458	4.9262	59.11	56.5403	51.1629
A4-R30	4	30	MEASURED	215.4053	306.960563	129.5157	-172.1195	18.1668	218.0	39.8484	28.3019
A4-R31	4	31	DERIVED_COMPLETE	218.9736	316.357243	158.4618	-151.1268	10.4325	125.19	35.7571	21.4575
A4-R32	4	32	MEASURED	223.2941	333.253754	199.4035	-100.4913	3.3333	40.0	65.1167	58.2338
A4-R33	4	33	MEASURED	226.5686	350.159794	223.2354	-38.7208	38.4746	461.7	66.2084	45.3044

Circle	Arm	Slot	Status	r ft	Theta deg	X ft	Y ft	d ft	d in	Spacing	Gap
A4-R34	4	34	MEASURED	229.3038	325.627351	189.2635	-129.4588	8.389	100.67	96.889	73.4572
A4-R35	4	35	MEASURED	232.2362	313.560055	160.0374	-168.2906	3.3333	40.0	48.6012	42.74
A4-R36	4	36	MEASURED	235.5089	315.524085	168.0462	-164.9998	12.7511	153.01	8.6586	0.6164
A4-R37	4	37	MEASURED	242.7561	326.097368	201.4843	-135.4053	10.5655	126.79	44.6535	32.9952
A4-R38	4	38	DERIVED_COMPLETE	244.237	340.725337	230.5468	-80.6219	6.366	76.39	62.0149	53.5491
A4-R39	4	39	MEASURED	245.6045	349.800482	241.7232	-43.4908	3.4833	41.8	38.7767	33.8521
A4-R40	4	40	MEASURED	250.4644	298.90317	121.0572	-219.266	70.8238	849.89	213.207	176.0535
A4-R41	4	41	MEASURED	251.9819	326.895993	211.0804	-137.6226	10.0956	121.15	121.5312	81.0715
A4-R42	4	42	MEASURED	252.0539	349.864164	248.1202	-44.3571	5.1666	62.0	100.3514	92.7203
A4-R43	4	43	MEASURED	253.0789	334.191749	227.8358	-110.1806	7.9432	95.32	68.8781	62.3232
A4-R44	4	44	MEASURED	254.3803	342.916954	234.1571	-74.7261	9.6029	115.23	38.6233	29.8503
A4-R45	4	45	DERIVED_COMPLETE	255.8598	338.804793	238.5519	-92.5052	7.7158	92.59	18.3658	9.7065
A4-R46	4	46	MEASURED	258.172	334.692633	233.3946	-110.3618	5.8287	69.94	18.5865	11.8142
A4-R47	4	47	MEASURED	265.0932	354.733333	263.9741	-24.3332	11.6574	139.89	91.3018	82.5588
A4-R48	4	48	MEASURED	271.9506	349.726518	267.5906	-48.5015	29.7616	357.14	24.4374	3.7279
A4-R49	4	49	MEASURED	274.1978	313.62672	189.1847	-198.4781	3.3333	40.0	169.2349	152.6875
A4-R50	4	50	MEASURED	275.1166	322.775962	219.0688	-166.4272	39.716	476.59	43.8215	22.2968
A4-R51	4	51	DERIVED_COMPLETE	278.1155	334.74548	251.5336	-118.6552	36.6359	439.63	57.7592	19.5833
A4-R52	4	52	MEASURED	280.4717	342.036514	266.7996	-86.5005	33.5558	402.67	35.5946	0.4988
A4-R53	4	53	MEASURED	280.4785	310.988586	183.9683	-211.7165	64.8123	777.75	150.1335	100.9495
A4-R54	4	54	MEASURED	285.847	299.415191	140.3894	-248.9967	3.3333	40.0	57.3492	23.2764
A4-R55	4	55	MEASURED	286.6087	334.268478	258.1881	-124.4325	43.2269	518.72	171.4431	148.163
A4-R56	4	56	MEASURED	290.0111	326.122132	240.7752	-161.6593	3.8158	45.79	41.0979	17.5766
A4-R57	4	57	MEASURED	294.6664	350.3559	290.502	-49.3648	6.7903	81.48	122.8121	117.509
A4-R58	4	58	DERIVED_COMPLETE	298.6306	350.488716	294.5254	-49.3463	6.6066	79.28	4.0234	-2.6751
A4-R59	4	59	MEASURED	301.54	350.526172	297.4273	-49.6326	6.4229	77.07	2.916	-3.5987
A4-R60	4	60	MEASURED	302.9511	325.139609	248.5857	-173.1604	30.4866	365.84	132.8331	114.3783
A4-R61	4	61	MEASURED	304.7024	343.342163	291.9151	-87.3446	3.3333	40.0	96.1342	79.2243
A4-R62	4	62	MEASURED	307.675	308.599551	191.9503	-240.4558	3.3333	40.0	182.8551	179.5218
A4-R63	4	63	MEASURED	313.4465	343.379321	300.3505	-89.6564	3.8158	45.79	185.7177	182.1432
A4-R64	4	64	MEASURED	320.2016	335.393291	291.1233	-133.3279	10.45	125.4	44.6356	37.5027
A4-R65	4	65	DERIVED_COMPLETE	325.3286	320.388419	250.6281	-207.4229	17.6413	211.7	84.439	70.3933
A4-R66	4	66	MEASURED	329.4791	310.155176	212.4679	-251.8211	21.7534	261.04	58.544	38.8467
A4-R67	4	67	MEASURED	332.6147	323.116065	266.0428	-199.634	17.2767	207.32	74.7915	55.2765
A4-R68	4	68	MEASURED	348.4342	311.767634	232.0959	-259.8805	7.789	93.47	69.1522	56.6194
A5-R01	5	1	MEASURED	40.0886	45.28498	28.2056	28.4875	6.5046	78.06		
A5-R02	5	2	MEASURED	46.5092	35.119183	38.0425	26.7558	7.5659	90.79	9.9883	2.953
A5-R03	5	3	MEASURED	65.6443	31.684297	55.8604	34.4789	24.1327	289.59	19.4196	3.5703
A5-R04	5	4	MEASURED	67.7113	12.383602	66.136	14.5211	4.9888	59.87	22.4478	7.887
A5-R05	5	5	MEASURED	68.517	51.031866	43.0895	53.2717	7.0553	84.66	45.086	39.0639
A5-R06	5	6	MEASURED	96.203	48.202301	64.1195	71.7196	3.3333	40.0	27.9747	22.7804
A5-R07	5	7	MEASURED	96.2119	31.433202	82.0927	50.1749	26.3511	316.21	28.0572	13.215
A5-R08	5	8	MEASURED	99.0439	39.379066	76.5575	62.8382	3.3333	40.0	13.8202	-1.022
A5-R09	5	9	MEASURED	99.8333	45.555177	69.9055	71.2735	3.5276	42.33	10.7426	7.3122
A5-R10	5	10	MEASURED	108.5564	19.002477	102.6406	35.3469	4.9888	59.87	48.6035	44.3453

Circle	Arm	Slot	Status	r ft	Theta deg	X ft	Y ft	d ft	d in	Spacing	Gap
A5-R11	5	11	MEASURED	113.6571	18.393879	107.8503	35.8642	3.8643	46.37	5.2354	0.8088
A5-R12	5	12	DERIVED_COMPLETE	118.6625	22.930011	109.286	46.2317	3.3667	40.4	10.4664	6.8509
A5-R13	5	13	MEASURED	122.6636	32.525041	103.4246	65.9523	3.3333	40.0	20.5733	17.2233
A5-R14	5	14	MEASURED	124.8218	47.985907	83.5449	92.7401	5.2323	62.79	33.3585	29.0757
A5-R15	5	15	MEASURED	128.5244	46.046225	89.206	92.5247	5.465	65.58	5.6652	0.3165
A5-R16	5	16	MEASURED	138.1066	36.106958	111.5789	81.3855	37.8953	454.74	24.9926	3.3125
A5-R17	5	17	MEASURED	141.8521	358.145734	141.7778	-4.59	3.5276	42.33	91.1249	70.4135
A5-R18	5	18	MEASURED	146.3826	358.681531	146.3438	-3.3682	4.7328	56.79	4.7266	0.5964
A5-R19	5	19	MEASURED	149.2703	38.713032	116.4739	93.3567	4.9888	59.87	101.232	96.3712
A5-R20	5	20	MEASURED	156.7947	26.815431	139.9337	70.7329	7.3996	88.8	32.5914	26.3972
A5-R21	5	21	MEASURED	160.8663	39.959462	123.3038	103.3156	4.4621	53.55	36.5812	30.6504
A5-R22	5	22	MEASURED	167.2115	54.369711	97.4095	135.9083	7.2295	86.75	41.6269	35.7811
A5-R23	5	23	MEASURED	172.0895	51.855038	106.2916	135.3399	8.9243	107.09	8.9003	0.8234
A5-R24	5	24	MEASURED	174.6593	5.327941	173.9047	16.2182	4.1739	50.09	136.9726	130.4235
A5-R25	5	25	MEASURED	179.3113	5.193398	178.5752	16.2309	3.3333	40.0	4.6706	0.917
A5-R26	5	26	MEASURED	180.1418	42.701758	132.3851	122.169	43.2044	518.45	115.5699	92.3011
A5-R27	5	27	MEASURED	189.259	9.409849	186.7123	30.943	5.6881	68.26	106.1773	81.7311
A5-R28	5	28	MEASURED	190.4682	1.622791	190.3919	5.3939	14.1984	170.38	25.8127	15.8694
A5-R29	5	29	MEASURED	193.8925	9.188833	191.4043	30.9625	3.8643	46.37	25.5886	16.5572
A5-R30	5	30	MEASURED	200.8418	34.259752	165.9946	113.063	9.5962	115.15	85.9428	79.2125
A5-R31	5	31	MEASURED	203.8656	8.745584	201.4953	30.9972	7.888	94.66	89.4153	80.6732
A5-R32	5	32	MEASURED	212.4226	8.174821	210.2641	30.2052	8.6409	103.69	8.8045	0.5401
A5-R33	5	33	MEASURED	213.9624	50.412765	136.348	164.8912	45.7232	548.68	153.6357	126.4536
A5-R34	5	34	MEASURED	220.6448	54.943328	126.7353	180.6163	3.3333	40.0	18.4305	-6.0978
A5-R35	5	35	DERIVED_COMPLETE	224.1409	38.431956	175.5801	139.3226	3.8087	45.7	63.9608	60.3898
A5-R36	5	36	MEASURED	226.604	21.920584	210.2211	84.5961	4.4621	53.55	64.7687	60.6333
A5-R37	5	37	MEASURED	228.4707	53.382167	136.277	183.3779	3.3333	40.0	123.3919	119.4942
A5-R38	5	38	MEASURED	228.6439	54.703575	132.112	186.6132	6.11	73.32	5.2739	0.5523
A5-R39	5	39	MEASURED	236.7399	22.328574	218.9892	89.9416	5.2323	62.79	129.9732	124.3021
A5-R40	5	40	MEASURED	244.5406	21.047881	228.225	87.8263	4.7328	56.79	9.4749	4.4924
A5-R41	5	41	MEASURED	246.2481	2.476534	246.0181	10.6404	56.3537	676.24	79.2102	48.6669
A5-R42	5	42	MEASURED	249.2638	49.698171	161.2273	190.1004	3.3333	40.0	198.4827	168.6392
A5-R43	5	43	MEASURED	249.6037	32.111502	211.4181	132.6815	3.8643	46.37	76.263	72.6642
A5-R44	5	44	MEASURED	253.7168	17.300799	242.2378	75.4524	21.283	255.4	65.0002	52.4265
A5-R45	5	45	MEASURED	257.3205	30.488099	221.742	130.554	3.5276	42.33	58.79	46.3847
A5-R46	5	46	MEASURED	263.6796	54.357201	153.6541	214.2834	3.8643	46.37	107.9193	104.2234
A5-R47	5	47	MEASURED	264.8623	5.972421	263.4246	27.5589	3.3333	40.0	216.6002	213.0014
A5-R48	5	48	MEASURED	266.4848	53.494237	158.5328	214.1997	3.8643	46.37	214.096	210.4972
A5-R49	5	49	MEASURED	266.7775	51.190187	167.1994	207.8812	11.4851	137.82	10.7254	3.0507
A5-R50	5	50	MEASURED	268.7532	15.851488	258.5333	73.4085	6.11	73.32	162.557	153.7594
A5-R51	5	51	MEASURED	273.0744	5.396168	271.8642	25.6804	3.3333	40.0	49.5549	44.8333
A5-R52	5	52	MEASURED	275.0448	48.638689	181.751	206.4369	12.3215	147.86	201.9735	194.1461
A5-R53	5	53	MEASURED	277.4516	43.549953	201.0897	191.1605	22.0864	265.04	24.6446	7.4406
A5-R54	5	54	MEASURED	281.9351	35.569795	229.3281	164.0	31.7486	380.98	39.1803	12.2628
A5-R55	5	55	MEASURED	282.1161	26.571517	252.3181	126.1946	29.6824	356.19	44.2469	13.5314

Circle	Arm	Slot	Status	r ft	Theta deg	X ft	Y ft	d ft	d in	Spacing	Gap
A5-R56	5	56	MEASURED	286.1417	13.566911	278.1573	67.1233	25.633	307.6	64.4755	36.8178
A5-R57	5	57	DERIVED_COMPLETE	288.2223	5.096502	287.0828	25.6038	19.6099	235.32	42.4681	19.8466
A5-R58	5	58	MEASURED	289.6865	1.027262	289.6399	5.1935	12.9132	154.96	20.5698	4.3083
A5-R59	5	59	MEASURED	290.8073	50.823449	183.7065	225.4347	6.8766	82.52	244.3933	234.4984
A5-R60	5	60	MEASURED	297.1806	44.979125	210.215	210.0619	3.3333	40.0	30.6435	25.5386
A5-R61	5	61	MEASURED	301.2801	44.178603	216.0693	209.9613	3.3333	40.0	5.8552	2.5219
A5-R62	5	62	MEASURED	301.4772	3.057104	301.0481	16.0781	3.3333	40.0	211.6886	208.3553
A5-R63	5	63	MEASURED	307.4462	35.515801	250.2475	178.6039	3.3333	40.0	170.2802	166.9469
A5-R64	5	64	MEASURED	308.2139	24.241075	281.0375	126.5455	4.9888	59.87	60.4823	56.3212
A5-R65	5	65	MEASURED	312.4493	34.741924	256.7481	178.0589	3.3333	40.0	56.9526	52.7916
A5-R66	5	66	MEASURED	314.731	11.987144	307.8681	65.3672	4.9888	59.87	123.7444	119.5833
A5-R67	5	67	MEASURED	316.5124	23.145593	291.036	124.4112	7.2295	86.75	61.3964	55.2872
A5-R68	5	68	MEASURED	325.3827	12.495259	317.6757	70.3994	3.8643	46.37	60.2241	54.6772